COMPARATIVE STUDY ON CONTROVERSIAL REDDIT POST PREDICTION USING DIFFERENT MACHINE LEARNING AND DATA MINING TECHNIQUES

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DHAKA, BANGLADESH

AUGUST 2023

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APPROVAL

This Research-Based project titled is **"Comparative Study on Controversial Reddit Post Prediction Using Different Machine Learning and Data mining Techniques"**, submitted by **Md. Rafeul Islam Rafe, ID No: 192-15-13264,** to the Department of Computer Science and Engineering, Daffodil International University has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on 5-August-2023.

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Hereby declare that, this project has been done by me under the supervision of **Fahad Faisal, Assistant Professor, Department of Computer Science and Engineering** Daffodil International University. I also declare that neither this project nor any part of this project has been submitted elsewhere for award of any degree or diploma.

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ACKNOWLEDGEMENT

First, I express my heartiest thanks and gratefulness to almighty God for His divine blessing to makes me possible to complete the final year project/internship successfully.

I really grateful and wish my profound my indebtedness to **Fahad Faisal**, **Assistant Professor**, Department of CSE Daffodil International University, Dhaka. Deep Knowledge & keen interest of my supervisor in the field of "*Data Mining, Machine Learning (ML), Artificial Intelligence (AI)*" to carry out this project. His endless patience, scholarly guidance, continual encouragement, constant and energetic supervision, constructive criticism, valuable advice, reading many inferior drafts and correcting them at all stage have made it possible to complete this project.

I would like to express my heartiest gratitude to **Dr. Touhid Bhuiyan, Professor and Head, Department of CSE**, for his kind help to finish our project and also to other faculty member and the staff of CSE department of Daffodil International University.

I would like to thank my entire course mate in Daffodil International University, who took part in this discuss while completing the course work.

Finally, I must acknowledge with due respect the constant support and patients of my parents.

ABSTRACT

Reddit has become one of the trendiest social media sites for expressing opinions and emotions. We can observe a steady shift in the pattern of Reddit posts, such as individuals starting to use slang, acronyms, and emoticons, making it difficult to determine the thoughts underlying the posts. We hope to present approaches, algorithms, and procedures involved in developing a Machine Learning model that can assess sentiments in this literature review. We utilized data directly collected from the Reddit post. For Sentiment Analysis, this study suggests using the KNeighbors Model, Naive Bayes Model, Decision tree Model, Random Forest Model, Random Forest Model, Logistic Regression Model, Linear SVC Model, MultinomialNB Model with Bidirectional LSTM (Bi-Lstm) is a recurrent neural network. By comparing with supervised machine learning model and recurrent neural network we found out supervised model Linear SVC provide the most accuracy, with 81% but compared with the complexity of the recurrent neural network, Bidirectional LSTM (Bi-Lstm) provide the best result at 78.21% also can predict sentiment with efficiency.

TABLE OF CONTENTS

CONTENTS	PAGE
Board of examiners	i
Declaration	ii
Acknowledgements	iii
Abstract	iv
List of Figure	viii
List of Tables	ix
CHAPTER 1: INTRODUCTION	**1-7**
1.1 Introduction	1
1.2 Objective	3
1.3 Motivation	4
1.4 Reason behind this study	5
1.5 Research questions	5
1.6 Research methodology	6
1.7 Expected outcome	6
1.8 Research layout	7
CHAPTER 2: BACKGROUND STUDIES	**8-16**
2.1 Introduction	8
2.2 Literature review	9
2.3 Summary of the research	14

2.4 Research Challenges 15

CHAPTER 3: RESEARCH METHODOLOGY 17-31

3.1 Introduction 17

3.2 Research subject and intermediary 17

3.3 Data collection procedure 18

3.4 Data fetching and data preprocessing 19

3.5 Arithmetical study 22

3.6 Visualizing data into word clouds 22

3.7 Executional requirement 24

3.8 Architecture of the model 26

CHAPTER 4: EXPERIMENTAL OUTCOME OVERVIEW 32-40

4.1 Introduction 32

4.2 Train and test split 32

4.3 Bidirectional LSTM with NN implementation 33

4.4 Implementation result 35

4.5 Descriptive analysis 39

4.6 Moral 39

4.7 Sentiment prediction 40

CHAPTER 5: IMPACT ON SOCIETY, ENVIRONMENT AND SUSTAINABILITY 41-42

5.1 Impact on society 41

5.2 Impact on student's life 41

5.3 Ethical issue 42

CHAPTER 6: CONCLUSION AND FUTUREWORK 43-44

REFERENCES 45-47

LIST OF FIGURES

FIGURES	PAGE NO
Figure 3.1.1 Steps of Data Collection	19
Figure 3.1.2 Clean Data Set Without Null	20
Figure 3.4.1 Words Counts	20
Figure 3.4.2 Word Length Variation	21
Figure 3.4.3 Clean text	21
Figure 3.5.1 Categorized Data	22
Figure 3.6.1 Word clouds	23
Figure 3.8.1 Workflow Model	26
Figure 3.8.2 Classification Models	27
Figure 3.8.3 Recurrent Neural Network	31
Figure 4.3.1 Plot Model Architecture	34
Figure 4.4.1 Confusion Matrix for The Passed Model and The Data	38
Figure 4.4.2 Representing Accuracy and Loss for Bidirectional LSTM	38
Figure 4.7.1 Sentiment Prediction	40

LIST OF TABLES

TABLES	PAGE NO
Table 1: Computer Hardware	18
Table 2: Accuracy of The Model	35-36
Table 3: Bidirectional LSTM Accuracy	37

CHAPTER 1
Introduction

1.1 Introduction

Recently social media connects all people worldwide and provides opinions or feelings on any topic. People from all around the world post on social media and write on different topics. Everyday people post millions of posts on social media. Software developers and requirements analysts must incorporate novel methods of bringing individuals from all over the world together as a result of the world's ongoing advancement in increasingly connected ways. On the other hand, users are getting more integrated with every business that collects and utilizes information. Developers are looking into social media platforms like Reddit to link people more actively and engage them in a variety of topics as a result of these parallel growths. Understanding how social media affects us might help us better understand its positive and negative effects. In order to better understand people's sentiment, we also look at how they express themselves on social media and compare that to other subreddit groups. Our analysis from this research project can be applied through mobile and online apps that implement machine learning and artificial intelligence. Social media sentiment affected trending topics, popular posts, and, most importantly, overall sentiment. Recently Reddit has become one of the most popular social media where people openly share all kinds of talks. Massive Reddit posts massive amounts of data generated every day. It generates a tremendous amount of structured and unstructured data. In recent years, social media has developed into a platform where people may express themselves. New developments in NLP are being made by studies that use data from social media sites to identify sentiment using learning techniques and neural networks. These posts are created by people with a wide range of ideologies and backgrounds. Statistics from the well-known network, Reddit, reveal that roughly 55.79 million people actively use it daily and that there are over 86400 posts made there. Many people can express their emotions in a virtual setting by using social media platforms. The large volume of text posts makes it easier to determine the signs of emotion and sentiment that a user is exhibiting. Simply put, sentiment analysis—also referred to as opinion mining—is the study of opinions or feelings drawn from textual data. There has been a lot of work done on this and there will be more in the future. For analyzing those data, sentiment analysis is applied to understand

post sentiment. Sentiment Analysis uses text classification used to analyze post text and classify for analyzing text and categorizes them as positive, negative, and neutral. Sentiment analysis systems are used for almost every social media platform, analysis of post opinions is central to almost all human activities and influences user behavior. Used a large amount of post data gathered from Reddit platform analysis with different techniques. Sentiment analysis considers a text classification task, a subfield of Natural Language Processing. Using NLP machine learning techniques to analyze, understand, and gain in-depth knowledge in human language with intelligence. We have attempted to develop a model that, in essence, recognizes the emotions concealed in posts by looking at the words used in the posts.

In this study, to determine sentiment, we used supervised machine learning classification methods KNeighbors, NaiveBayes, DecisionTree, RandomForest, Logistic Regression, Linear SVC, MultinomialNB. We decided to use bidirectional recurrent neural networks (biLSTM) in order to get better results. Bidirectional LSTMs can be made by extending traditional LSTMs, which can improve model performance for sequence classification problems. When all timesteps of the input sequence are available for a task, bidirectional LSTMs train two LSTMs rather than one on the input sequence. the first on a reversed copy of the input sequence, and the second on the original input sequence. This can help the network learn the issue more thoroughly and faster by providing it with more context. In a bidirectional LSTM, we provide input in both the right-to-left and left-to-right directions. Keep in mind that this is merely input coming from both sides; there is no backward propagation. Therefore, if we have two inputs, the question is how the data gets integrated in the output.

Normal RNN networks do not allow us to predict the following word, but bidirectional RNN networks can. RNN networks may also be LSTM or GRU. Sentiment analysis, text classification, text production, and machine translation are a few of the well-known applications that use BiLSTM.

The results from this can be used in a variety of situations, such as the analysis and monitoring of shifts in sentiment around an event, feelings toward a certain organization, or the launch of a particular product. This study emphasizes the value of sentiment analysis in comprehending human emotions, and I hope it encourages more research in this area. This paper implements both machine learning and deep learning techniques to classify text sentiment from text posts from Reddit. This

paper outline follows a brief discussion of relevant work on sentiment analysis in the "Background and literature review" section, the proposed work, and the technique for affect and sentiment analysis described in the 'Proposed approach' section. The 'Result and discussion' section discusses experimental results.

1.2 Objectives

There are various advantages of utilizing AI in social media. There are various technological and strategic goals for engaging more users with the platform.
Some of the technical objectives are listed here.:

- Create an effective model for detecting user sentimental behaviors.
- To encourage software developers to cooperate with AI by utilizing the model.
- Incorporate the model into mobile applications and websites.
- Assist the user in adhering to the platform's rules and regulations.
- Make the social media platform more user friendly.
- Investigate modern machine learning methods and neural network designs to improve the accuracy of sentiment analysis.
- Examine the efficacy of deep learning approaches in sentiment analysis on social media data, such as transformers and attention processes.
- Investigate the use of multimodal data (text, photos, and videos) in sentiment analysis to gain a more complete knowledge of user sentiment.
- Improve the model's capacity to interpret subtle sentiments by developing approaches for dealing with the issues of sarcasm, irony, and figurative language in sentiment analysis.
- Examine the influence of using domain-specific information and lexicons in sentiment analysis to increase contextual understanding of sentiment in certain sectors or domains.
- Develop strategies for real-time sentiment analysis to enable immediate detection and response to emerging sentiment trends on social media platforms.
- Evaluating the ethical implications and potential biases in sentiment analysis algorithms, as well as proposing techniques for minimizing prejudice and guaranteeing fairness in sentiment classification.

1.3 Motivation

In recent years, social media has undergone a worldwide revolution. The emergence of new forms of social media usability in the field of information and communication technology is noteworthy. Furthermore, social media is used to disseminate postings, thoughts, and remarks that may be offensive to others.

During the epidemic, sentiment analysis was used to determine how the world crisis affected people's emotions. I investigated the sentiments expressed on Reddit using sentiment analysis and discovered that they drastically varied in reaction to important subjects and events. This study emphasizes the value of sentiment analysis for comprehending human emotions, and I believe it will encourage more research in this area.

However, thanks to suitable processes and capacities of Artificial Intelligence and Machine Learning in this subject. We can overcome this type of difficulty by utilizing modern technologies. We need to invest more in social media platforms in this subject if we want to establish a smart social media platform that will engage people of all ages and backgrounds. Traditional approaches for detecting issues and determining potential remedies in this area are quite challenging. Furthermore, most do not address user experience; instead, they focus on increasing user engagement approaches.

We can identify users who publish in violation of the rules, figure them out, and observe their behavior. We can also filter out threats against persons, threats against minorities, and bad attitude. Sentiment analysis is the process of swiftly determining a text's sentiment, such as whether it is favorable or negative.

Artificial intelligence will assist us in determining the sentiment analysis of every post on social media sites. This thesis focuses on Reddit-based social media systems that incorporate artificial intelligence, machine learning, and neural networks. This can help in the detection of emotion in Reddit postings.

1.4 Reason behind this study

Sentiment analysis is the top research field under NLP. It utilizes textual data available almost on social media like Reddit to analyze people's opinions and concentrates on the subject part of the text that leads to a good or bad direction. We want to compare its effectiveness in detecting sentiment to traditional supervised learning in this study. Even though deep learning approaches are often used in this industry, we argue that simpler machine learning models, which use fewer computing resources, still have a role. conventional machine learning models, although there are several techniques and methodologies in the field of NLP, they are both inadequate and severely constrained. Therefore, the scope of NLP research has to be expanded. When working with a lot of trash and null posts in other languages, preprocessing is the main problem. The anticipated results, As the outcome, in our study project we are striving to analyze text and describe how to do so. Our goal is to produce a text analysis. This enables us to provide the most accurate sentiment analysis summaries and to summarize text.

As a user, anyone can get offended by other users' opinions on any bad talk. We need to identify these posts or write for social media to filter those posts. On the other hand, it helps identify hate speech or any crime speech on Reddit to decide what action we should take for any post or opinion. To give a proper solution it’s necessary to find out the problems and related requirements in this field.

1.5 Research Questions

Below are some of the key questions that this research paper is focused on:

- How is sentiment analysis doing right now in the world of social media?
- What are the restrictions on using sentiment analysis with artificial intelligence, machine learning, and neural networks?
- How may the limits of sentiment analysis be overcome?
- What is sentiment analysis, exactly?
- What is the text analysis method?
- What does sentiment analysis mean?
- What is the definition of sentiment analysis?

- What advantages can sentiment analysis offer?
- How can I gather data for projects by using web scraping?
- What is the process of a machine learning model?
- How can data collecting be made better?
- How do neural network's function?
- How should sentiment be classified?
- Which approach to sentiment analysis is the best?
- What procedure is used for text cleaning?
- What can sentiment analysis serve as a tool for?
- What are the difficulties that this research project is facing?

1.6 Research Methodology

In this section We describe dataset, data pre-processing, model architecture, machine learning models, and model training in this section of our research study. The performance of the suggested model will be discussed at the conclusion of this chapter.

1.7 Expected Outcome

Our main goal was to identify sentiment from any post or sentence posted by any user on a social media platform. The study can provide an indication of positive, negative, and neutral attitudes expressed in the postings, giving a general picture of sentiment distribution. This data aids in comprehending the general sentiment landscape surrounding a certain subject or business. Social media influencers or online profiles that promote either good or negative sentiment can be found using sentiment analysis. This data helps businesses find prospective brand ambassadors or lessen the influence of unfavorable influencers. Sentiment research may show that the majority of social media messages are favorable. This might imply that individuals are sharing pleasant experiences, expressing joy, or promoting positive messages or items. Sentiment analysis, on the other hand, may suggest a prevalence of negative sentiment in social media messages. This might imply discontent, annoyance, criticism, or unfavorable reactions to a certain topic, event, or product. Several entries may have a neutral attitude, which is they do not express significantly positive or negative sentiments. These postings may be informative, factual, or devoid of emotional content.

1.8 Research Layout

Chapter 1: will provide an overview about the introduction, motivation, problem definition, research question, research methodology, and expected outcome of our project.
Chapter 2: will cover the history of this research, related work, and present status using the Reddit social media platform.
Chapter 3: will describe situation of AI in the field of agriculture in Bangladesh.
Chapter 4: will discuss about development perspective for the AI in agriculture.
Chapter 5: it is focus to the result and benefit of using AI in agriculture.
Chapter 6: It describes the conclusion and findings of this study.
Chapter 7: All of the references we utilized for this research are listed below.

CHAPTER 2
Background Studies

2.1 Introduction

Considering the purpose of this research is to investigate supervised ways to determine the sentiment of Reddit posts, only studies relevant to analyzing the sentiment using these approaches are examined in this section. Pre-processing procedures, feature extraction methods, suggested methodology (classification methods), and assessment metrics are all discussed in these studies. Sentiment analysis, often known as opinion mining, has grown as an important study subject within natural language processing (NLP). Individuals now have an unparalleled capacity to share their ideas and feelings on a wide range of issues because to the increasing usage of social media platforms. Understanding and evaluating these feelings has become critical for businesses, organizations, and academics seeking to obtain insights into public perception, consumer feedback, and new trends.

Sentiment analysis has advanced significantly in recent years, owing to the increased availability of large-scale textual data and the development of powerful machine learning and deep learning algorithms. Researchers have investigated a variety of methodologies, including classic supervised learning algorithms as well as more sophisticated methods such as recurrent neural networks (RNNs) and transformers.

Numerous researches have focused on sentiment analysis utilizing social media data, notably from sites such as Twitter, Facebook, and Reddit. These platforms provide extensive sources of user-generated material that may be used to derive useful insights regarding public opinion toward products, services, events, and social concerns. To assess sentiment patterns and classify material into positive, negative, or neutral categories, researchers used a variety of methodologies such as text processing techniques, feature extraction methods, and machine learning models.

In this thesis article, we intend to add to the field of sentiment analysis by studying and analyzing several supervised machine learning algorithms and recurrent neural network architectures for sentiment categorization. We will train and evaluate our models using real-world social media

data, especially from the Reddit platform. In addition, we will look into approaches to improve the accuracy and interpretability of sentiment analysis results, taking into account the issues given by user-generated content on social media.

2.2 Literature Review

This paper is to study Reddit social media posts with supervised approaches for sentiment analysis. Related research analyzes proposed work methods, feature extraction methods, data-preprocessing, and evaluation. Research [1] work Proposed an aspect of text review, analyzing these reviews as positive or negative with a machine learning approach consisting of supervised and unsupervised algorithms that provide a reliable result of sentiment analysis. Social media sentiment analysis and opinion mining to view popular public opinion analysis. This paper [2] provides outline studies on Facebook and Twitter. It provides direction work with sentiment analysis in work. Paper [3] on deep learning-based sentiment analysis and sentence level extract the topic by using an extended short-term memory network to perform sentiment analysis. The proposed work [4] with CNN and Bi-LSTM uses classification semantics with contextual information and performance compared with baseline methods. This study work [5] used machine learning approaches like SVM, logistic regression, and random forest to provide results that perform better for classification. This work combines deep learning with Amazon product reviews in 12 categories of Amazon products. Research article [7] Researchers from a variety of institutions, including universities, the government, and business industries, are interested in sentiment analysis from internet evaluations. To achieve outstanding results, deep learning (DL) algorithms are used in sentiment analysis. For sentiment analysis, this paper presented a hybrid convolutional neural network-long short-term memory (CNN-LSTM) model. To obtain results, our suggested model is implemented using dropout, max pooling, and batch normalization. The term "sentiment analysis" (SA) is gaining popularity and becoming trendy. Recent advances in machine learning are used by MSA to advance. At each stage of the MSA, the most recent developments in machine learning and deep learning are used, including sentiment polarity recognition, multimodal features extraction, and multimodal fusion with reduced error rates and increased speed [8]. [9] Online retailers like Amazon offer a webpage where customers may voice their opinions about various products. To extract relevant information, such as positivity or negativity, the system executes pre-processing operations like stone-coating, tokenization, boxing,

and elimination of stop-words from the datasets. Last but not least, they provide some insight into their upcoming text classification work. In this paper [10] After pre-processing, the collected data is used to create the supervised machine learning model using the relevant raw data that was obtained from the Internet Movie Database (IMDb) website. The results of the hybrid model, which combines the two models, are optimistic for success prediction ratings for new films. The top movies of a specific year can be found using the proposed approach. Consumers can offer precise and impartial sentiment suggestions by using aspect-based sentiment analysis (ABSA). To start, we firstly suggest a deep-level semi-self-help sentiment annotation system based on the weakly supervised bidirectional encoder representation from transformers (BERT) classifier [11]. The accuracy of the ABSA tasks can be effectively increased, and the space-time complexity can be decreased, using the suggested approach. This article [12] The impact of reviews on the sales of mobile phones is examined in this paper. This study investigates how the opinions of various mobile phone companies in online reviews affect their sales, taking into account the conflicting results on the effect of reviews in earlier work. In this research, we examine comments and reviews on smartphones from Amazon as a data set, classifying the review content as neutral, highly neutral, or unfavorable. Comparative tests have been performed between our model and sentiment categorization. [13] used social cognition theory in an effort to infer and extract the emotion notion from each emotional review. In this work, we used a sample of 4,300 comments from dating websites with extremely negative or positive emotions and three machine learning algorithms to identify the feelings. Discovered that lexicon-based methods combined with machine learning can reach higher accuracy than any kind of sentiment analysis.[14] As opposed to sentimental analysis on the review text, which aims to understand the overall emotion of the reviews as has been done in the past, the major work will involve empirical analysis for data understanding and exploration by taking into consideration, the various metrics related to the user reviews. From the paper [15] Three classification methods are used in this study: Support Vector Model (SVM), Logistic Regression, and Nave Bayes. Data retrieval, transformation, classification training, test data prediction, and outcome analysis were the steps in this study. The outcome of this research will depend on how the majority of Twitter users perceive the functionality of this mobile travel app. [16] Using a recurrent neural network to classify sentiment. This is the stage where we analyze, gather, display, and summarize data for subsequent processing. In the "modern" world of today, where ambiguous tweets are common, the trained model performs considerably more correctly

and with a lower margin of error. more so with RNN. In order to ascertain what emotional manifestations happened on a specific topic in a certain time period, we utilize this recently scraped data sets (by the keyword's theme) with our RNN model that we have constructed and trained. An overview of research [17] that attempted to determine the political attitude of internet users who used social media sites like Facebook and Twitter is also included in this essay. Additionally, this report shows ongoing problems with sentiment analysis research as well as difficulties in foretelling election results. Additionally, this study makes some recommendations for the future of using social media material to predict particular elections. This research [18] suggests a topic-level sentiment analysis algorithm that is based on deep learning. The proposed method is new because it uses online latent semantic indexing with regularization constraints to extract the subject at the sentence level before using topic-level attention mechanisms in long short-term memory networks to conduct sentiment analysis. The suggested approach is distinctive in that it provides topic modeling that is both scalable and dynamic across streaming short text input as well as topic-level sentiment analysis. From work [19] a unique automated methodology for a successful online product sentiment analysis is suggested in this paper. For a better comprehension of the multimodal data gathered from the Amazon websites, picture and data normalization techniques are used. Additionally, for the purpose of extracting the textual and visual feature vectors from the preprocessed data, the following descriptors are used: Latent Semantic Analysis (LSA), Term Frequency- Inverse Document Frequency (TF-IDF), Modified Local Binary Pattern (MLBP), and Speeded Up Robust Features (SURF). Finally, feature vector optimization and sentiment classification are combined using the Random Evolutionary Whale Optimization Algorithm (REWOA) and Deep Belief Network (DBN) classifier. Paper work [20] The aim of the study is to examine the emotions of people who tweet their comments using the hashtag #Monkeypox in a good, negative, and neutral manner. A hybrid-based model architecture built on CNN and LSTM was utilized to assess the prediction accuracy in order to learn what individuals believe about this illness. The outcome of the forecast using the whole set of monkeypox data.[21] The sentiment analysis approach presented in this study is based on Hindi. Due to the complexity of Hindi sentiment analysis, an LSTM model combined with a continuous bag of words model is employed. To improve accuracy, the input vector's ideal word size is calculated. The projected outcomes are contrasted with state-of-the-art techniques. It has been determined that the recommended approach is superior to the methods that were picked. This study [22] looks at the part that Reddit, a social

media site, played in the early 2021 events surrounding the GameStop (GME) share rally. We focus on how talks on the r/WallStreetBets subreddit have affected the pricing fluctuations at the American online store GameStop. Based on the Valence Aware Dictionary and Sentiment Reasoner (VADER) sentiment analysis tool, we adapt a sentiment analysis dictionary for Reddit platform users and analyze text from 10.8 million comments. An examination of the connections between Reddit sentiments. This article [23] suggests a unique unsupervised sentiment analysis technique based on lexicons to gauge "hope" and "fear" around the Ukrainian-Russian Conflict in 2022. Reddit.com the results of an experimental study demonstrate that following the symbolic and tactical defeats of Azovstal (Mariupol) and Severodonetsk, hope is significantly reduced. After various non-military events, like the Eurovision Song Contest and football games, there are spikes in optimism and anxiety, both positive and negative. Authors [24] want to present the approaches, algorithms, and process used in developing a Machine Learning model that can assess sentiments in this literature review. The "sentiment140 dataset" AND "TWITTER AND REDDIT SENTIMENT ANALYSIS DATASET" were the sources of the datasets we utilized. The usage of the Logistic Regression Model, Linear SVC Model, and Bernoulli NB Model for Sentiment Analysis is suggested in this study. From this [25] study creates word/sentence data that can be fed to machine learning models using a variety of NLP (Natural Language Processing) tools, including word embeddings from ELMo (Embeddings from Language Models), tokenizers from BERT (Bidirectional Encoder Representations from Transformers), and the BoW (Bag of Words) approach. Each method's outcomes have been explored. The findings of this study are noteworthy and may be used to examine mental stress among social media users in practical contexts. Although the emphasis of this study is Reddit data, the methods employed may be applied to other social media platforms and may aid in addressing the rising mental health epidemic.[26] These methods are employed in this study to systematically find, gather, measure, and analysis emotional states and private information in tweets and posts on Reddit and Twitter. Our main objective is to identify depression in these articles because it is a serious public health issue. We utilize a combined collection of Twitter and Reddit data to do this. Our study intends to further the growing body of research on understanding and predicting mental health consequences using social media data.

[27] For instance, by looking at users' posts on Reddit (a prominent message board), suicide intent might be detected. To examine several machine learning classification models, including Logistic Regression, Support Vector Machine (SVM), Random Forest, AdaBoost, XGBoost, and Gradient

Boosting, together with feature extraction methods like TF-IDF, Glove, or Elmo, is the goal of the current study. In our early tests, we discovered that the TF-IDF with SVM model performed the best, with an accuracy of 91.50% when applied to the SuicideWatch dataset. Our models were evaluated using an external dataset that had not been utilized in the training phase, nevertheless, to prevent knowledge leaking in order to generalize the outcomes of our models. This paper [28] In this work, machine learning techniques were employed to automate the detection of hate speech from the literature. In this study, we apply topic modeling to examine the content and identify subjects from comments designated as Islamophobic. Deep learning techniques are used to detect Islamophobia through Reddit. The Islamic dress code, religious customs, marriage, and politics are a few of the themes we mentioned. Using deep learning models, we were able to identify Islamophobia. From this study [29] The testing process on the previously created dataset is expanded in this work employing deep learning methods and a varied collection of input characteristics, including both hand-crafted and raw text features. The multilingual state-of-the-art model XLM-RoBERTa beat other algorithms utilizing raw untranslated text in experiments, outperforming standard techniques and scoring higher. using both conventional and deep learning methods together. This research work [30] In order to identify the category to which the ciphertext belongs, this research provides an effective network model based on deep BLSTM-GRU. One-dimensional convolutional neural network (1D-CNN) and various other deep learning-based approaches were used to compare the performance of the proposed model against two publicly available datasets encrypted with different classical cipher methods.[31] In this study, we present a natural language processing (NLP) method based on the Bidirectional Long Short-Term Memory (Bi-LSTM) methodology to carry out sentiment classification and identify different problems relating to COVID-19 public views. An enhanced form of traditional LSTMs called Bi-LSTMs may provide output from both the left and right contexts at each time step. When compared to the standard LSTM model and more recent research that are currently accessible in the literature, our experimental results using real datasets taken from Twitter and Reddit social media sites demonstrated superior metrics.[32] There are many different types of responses on Twitter that take the form of tweets. It is critical in today's society to categorize the many sentiments expressed in tweets. In this paper, a bidirectional LSTM implemented sentiment analysis of tweets based on natural language processing (NLP) has been developed. This research uses sentiment analysis from microblogging to examine how consumers feel about a company's products. This research work

[33] Building with privacy in mind is more crucial in this endeavor because of the growing user expectation for personal information. In order to better understand developers' opinions and problems while creating apps in the present privacy-focused society, we analyze the worries, issues, and solutions they debate on Reddit forums in this research. While there are common patterns in privacy concerns throughout the many subreddits, the frequency of such subjects varies between online and mobile applications, according to various kinds of Natural Language Processing (NLP) that we undertake.

2.3 Summary of the Research

The assessment of the literature indicates a varied spectrum of research papers focusing on sentiment analysis and related subjects in the context of social media platforms such as Reddit and Twitter. These studies use a variety of methodologies, such as supervised machine learning algorithms, deep learning techniques, and natural language processing (NLP) approaches, to assess sentiment and extract important insights from user-generated material.

Some studies look at sentiment analysis in a broad sense, examining the overall sentiment of posts or comments on social media networks. They use techniques such as K-Nearest Neighbors, Naive Bayes, Decision Trees, Random Forest, Logistic Regression, Linear SVC, and MultinomialNB to classify sentiment with varied degrees of accuracy. Furthermore, researchers use recurrent neural networks (RNNs), notably Bidirectional LSTM models, to collect contextual information and improve sentiment categorization performance. Several studies concentrate on specific areas of sentiment analysis, such as aspect-based sentiment analysis (ABSA), which seeks to detect sentiments associated with certain components or features of a product or service. To accomplish accurate sentiment categorization, researchers employ techniques such as BERT classifiers and deep learning-based systems. The paper also emphasizes the use of sentiment analysis in a variety of sectors, such as movie reviews, mobile phone sales, emotional ratings on dating websites, hate speech identification, and the detection of feelings associated to specific events like as the COVID-19 epidemic. Another major topic of study is privacy issues and their consequences. Studies look at user perspectives and privacy concerns in app development, highlighting the importance of privacy-focused solutions. NLP algorithms are used to analyze Reddit forum discussions, revealing similar trends and differences in privacy issues between online and mobile apps. Overall, the assessment of literature illustrates the importance of sentiment analysis in understanding public

opinion, consumer sentiment, and the influence of evaluations across a variety of areas. It demonstrates the efficacy of various algorithms, deep learning models, and natural language processing techniques in sentiment analysis tasks. These research' findings contribute to the development of more accurate sentiment analysis methodologies, a better understanding of privacy problems, and the enhancement of sentiment analysis applications in practical scenarios.

2.4 Research Challenges

The study reported in this paper is fraught with difficulties, notably in the areas of data gathering and data cleansing. These difficulties may have an influence on the accuracy and dependability of the study findings. Some of the major problems are as follows:

Data Availability: Collecting a large volume of data from social media networks such as Reddit and Twitter might be difficult. Access to comprehensive and relevant information may be restricted, and obtaining a varied selection of posts or comments that represent different moods and perspectives can be time-consuming.

Data Bias: Because of user self-selection and the prevalence of bots or spam accounts, social media data may be inherently skewed. This bias has the potential to alter sentiment distribution and the generalizability of study findings. Researchers must exercise caution on the representativeness of the data obtained and investigate ways to reduce bias.

Noise and Irrelevant Information: Irrelevant or off-topic postings, spam, or duplicate material are common sources of noise on social networking sites. It is critical to clean the data and remove such noise to ensure the accuracy and reliability of sentiment analysis findings. It might be difficult to develop strong algorithms or manual annotation procedures to filter out noise.

Data Pre-processing: Pre-processing the acquired data entails numerous activities, such as deleting special characters, URLs, usernames, and stop words. It also contains tokenization, stemming, and lemmatization. The variety and diversity of social media data can make pre-processing difficult, since there may be changes in language, slang, acronyms, and informal phrases that must be accounted for.

Labeling & Annotation: Assigning sentiment labels to acquired data for training and assessment purposes necessitates human annotation or the usage of pre-labeled datasets. It can be difficult to ensure consistent and accurate categorization since various annotators may have subjective interpretations of sentiment. Setting explicit criteria and doing inter-annotator agreement checks can help alleviate this issue.

Privacy and Ethical Considerations: Analyzing social media data poses privacy concerns since it involves the processing of personal information as well as user-generated content. It might be difficult to ensure compliance with data protection standards, ethical norms, and gaining adequate user permission for data usage. Throughout the data collecting and analysis process, researchers must emphasize user privacy while adhering to ethical norms.

Addressing these problems necessitates careful planning, effective data gathering tactics, rigorous data cleaning and pre-processing procedures, and adherence to ethical principles. By overcoming these obstacles, the research endeavor may improve the quality and validity of its findings, resulting in more accurate sentiment analysis results and important insights into public opinion.

CHAPTER 3

Research Methodology

3.1 Introduction

Finding out the exact sentiment is a difficult task due to various factors. Texts need to be opinionated in sentence posts with huge sentences in one direction that can be positive or negative. Therefore, we use better methods for automatic feature selection and extend that to data pre-processing, word embedding, and training and testing post-review. In our research work, we mainly focus on Sentiment Analysis at the sentence level. This paper's research methodology section provides a full explanation of the methodologies and strategies used to carry out the sentiment analysis of social media data study. This section describes the research strategy, data gathering procedure, data preparation techniques, and several analytical approaches used for sentiment categorization. The goal of this study technique is to offer a clear knowledge of the strategy used to examine and evaluate sentiment trends in social media posts.

- Conceptualization of Research
- Data Gathering
- Preprocessing of Data
- Extraction of Features
- Sentiment Analysis
- Evaluation
- Ethical Issues

3.2 Research Subject and Intermediary

"Comparative Study on Controversial Reddit Post Prediction Using Different Machine Learning and Data Mining Techniques," according to the research topic. The research subject focuses on performing a comparison analysis to forecast contentious Reddit postings using various machine learning and data mining approaches. This research is especially relevant in the field of Natural Language Processing (NLP). It is critical to have access to powerful hardware, notably a high-

performance PC with a GPU, to ease the implementation of deep learning models. Given the hardware requirements, cloud computing platforms become a viable option. The Kaggle notebook platform was chosen for this study because it supports accelerated GPUs, allowing for quicker calculations and model training.

Required equipment: This research project requires the following equipment:

- High-Performance Computer
- Platform for cloud computing
- Data Mining Applications
- Frameworks for machine learning
- Text Summarization Resources

Table 1: Computer Hardware

Hardware and software	**Development Tools**
Intel Core i7 or more powerful processor	Windows 10 or Linux
1TB HDD or 500MB SSD	Python 3.11
Google Colab, Jupyter Notebook , Kaggle Notebook	TensorFlow, PyTorch, Scikit-learn, Keras, NLTK (Natural Language Toolkit), Pandas
RAM 12GB or more	GPU & TPUs are available on Kaggle

3.3 Data Collection Procedure

The data for this study was obtained directly from the Reddit website using figure 3.1.1 Python programming and the PRAW (Python Reddit API Wrapper) module. The Reddit API allows developers to access Reddit's massive quantity of publicly available data, such as posts, comments, user information, and other pertinent metadata.

3.4 Data fetching and data pre-processing

Data preparation is a powerful method for cleaning up our own dataset. We cannot attain the expected results without a decent dataset. Our collection includes several types of news data, including sports news, economic and financial news, political news, and news of various contents. Obtaining Bengali information has various disadvantages, including the way the content is arranged. Our dataset contains sorted and summarized text content.

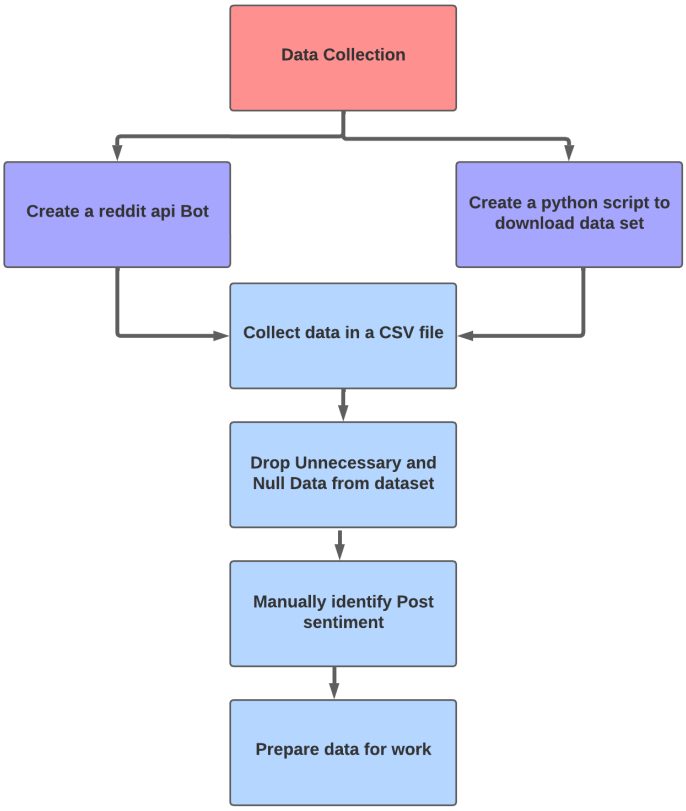


Figure 3.1.1 Steps of data collection

Our data was collected directly from the Reddit website using Reddit-API, PRAW, and Python script to create a bot that collects all top posts from the subreddit and combines and stores them in CSV format. Raw data directly collected from web sites has a lot of incorrect syntax and null values. Also, a few column values are null and in the wrong format. Raw data used processed with Excel and fixed data format, and columns remove null rows figure 3.1.2.

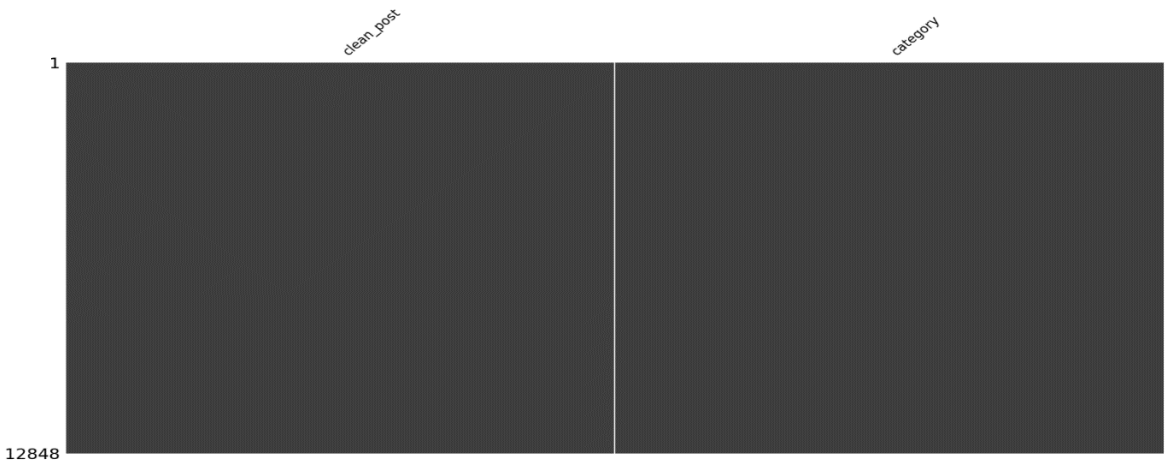


Figure 3.1.2 Clean data set without null

3.4.1. Conflicting issues

Because we used a massive dataset. So, there are a lot of words. We created a broad vocabulary such that every term is related to another. Long text yields a brief summary. Large datasets may reveal biases in the population from whom the data is drawn. These biases might provide biased or unjust findings. To maintain the validity and fairness of their studies, researchers must carefully examine and minimize any biases in the dataset.

3.4.2. Vocabulary counting

The model contains a lexical set, and the need for a vocabulary set to identify comparisons between the two descriptions and the summary is much more acceptable here. That is why we must count the vocabulary. Reddit posts, on average figure 3.4.1, were around 830 words long and upper to 7353 words.

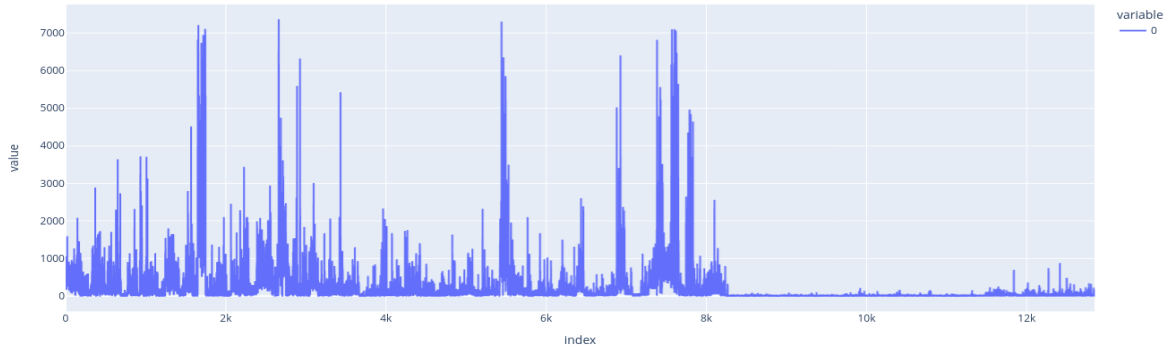


Figure 3.4.1 Words counts

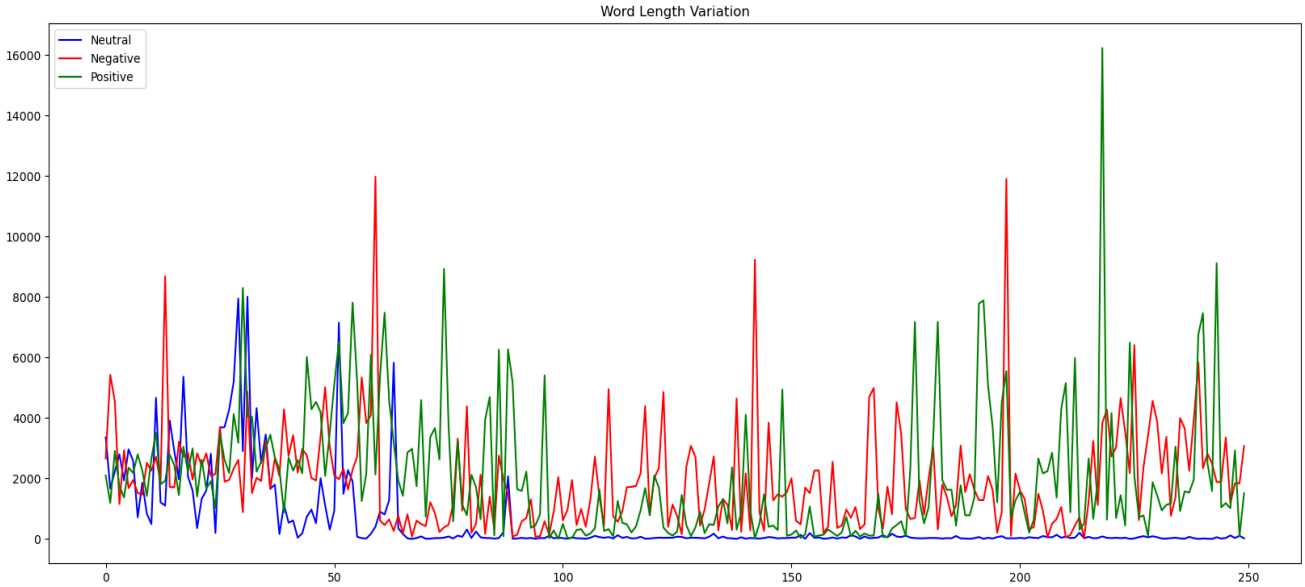


Figure 3.4.2 Word length variation

3.4.3. Cleaned-up text & summary

Text pre-processing is essential for improved accuracy. After processing, the text becomes lot cleaner and more readable. We eliminated the stop words, additional space, English letters, punctuation, and abbreviations during the pre-processing procedures. Text pre-processing figure 3.4.3 in the table as under,

Author	post_text	s
Roxy175	I had a group of queer friends before I know I...	
viantmoth	I'm gynesexual. I'm attracted to people who lo...	
LittleStick	I came out to my wife as bisexual this morning...	
grant_b48	Hi so I've been super curious lately and even ...	
Available4	Kristen Stewart,\nAnya Taylor Joy,\nDaisy Edga...	
...	...	
Roxy175	i had a group of queer friends before i know i...	
viantmoth	i'm gynesexual. i'm attracted to people who lo...	
LittleStick	i came out to my wife as bisexual this morning...	
grant_b48	hi so i've been super curious lately and even ...	
Available4	kristen stewart, anya taylor joy, daisy edgar-...	

Figure 3.4.3 Clean text

3.5 Arithmetical Study

The dataset utilized to classify sentiment analysis has two crucial aspects

- clean_post - actual post text data posted by Reddit users.
- Category - manually categorized data into 3-category (positive, negative, neutral) figure 3.5.1

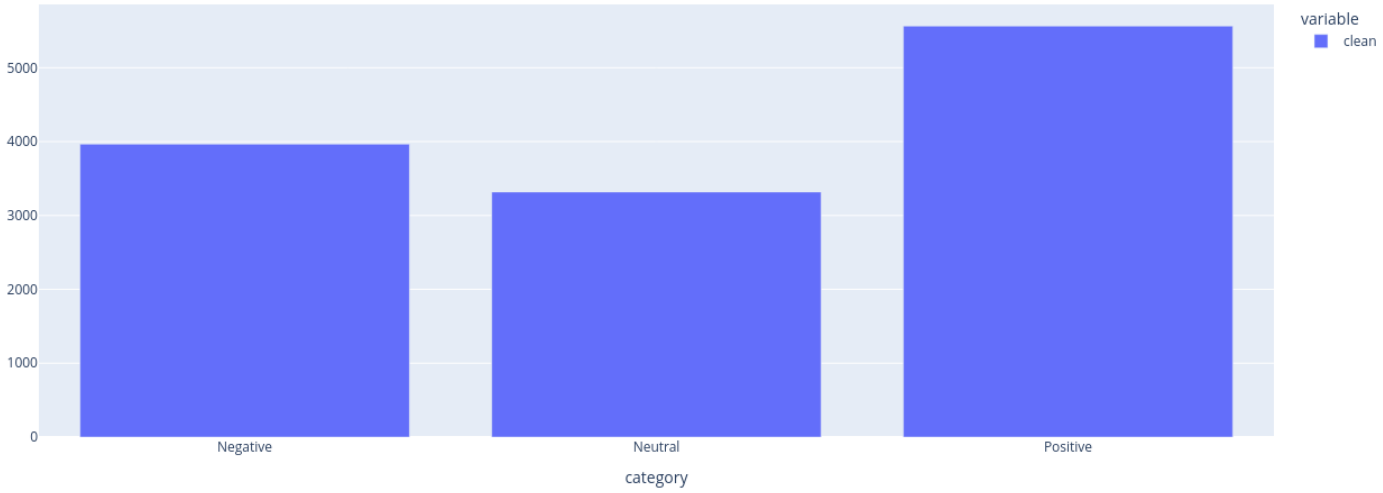


Figure 3.5.1 Categorized data

The dataset contained two columns, "clean_post" and "category." No null value was present in the data set, and the data was clean.

```
Index(['clean_post', 'category'], dtype='object')
```

Dataset "category" is categorized into three categories Positive, Negative, and Neutral.

- Positive - 43.3%
- Negative - 30.9%
- Neutral - 25.8%

3.6 Visualizing Data into Word Clouds

Word clouds are a common approach for providing a visual representation of text data. The magnitude of each word in a word cloud corresponds to its frequency or relevance in the dataset. This kind of visualization provides a quick and easy approach to find the most prevalent or

noteworthy phrases in the data. Word clouds may be constructed using numerous libraries and tools in computer languages such as Python. Popular Python packages, such as Word Cloud, offer simple methods for constructing word clouds figure 3.6.1. These libraries often include choices for customizing the look of the word cloud, such as font size, color palette, and word placement.



Figure 3.6.1 Word clouds

3.7 Executional Requirement

3.7.1. Problem discussion

The dataset's rundown and depiction inputs are both unbiased. A result or outline is usually more limited than a depiction (text). Consider W to be the actual word count of the dataset's feedback chain representation (text). The word size should be V, and the information chain should begin at 1. 2..3. The information chain, on the other hand, generates result chains such as y1, y2..., and y, where S>W. It implies that the rundown's chain isn't substantially lengthier than a text report. Words form each link in the chain.

3.7.2. Vocabulary and word embedding

The vocabulary in this research article refers to the collection of distinct words or terms found in the textual data utilized for analysis. The breadth and quality of the vocabulary are important factors in a variety of natural language processing (NLP) activities, such as sentiment analysis, text categorization, and topic modeling. The researchers plucked terms from the gathered dataset, which was taken straight from the Reddit website, to develop the vocabulary. The dataset was made up of Reddit user posts and comments, giving a varied range of language and subjects for examination. The vocabulary was created by aggregating all of the different terms in the dataset.
In contrast, word embedding is a technique for representing words as dense vectors in a continuous vector space. It captures the semantic links and contextual information of words, allowing for improved comprehension and analysis of textual material. The goal of the research article was to increase the knowledge of the dataset and the performance of NLP tasks by using word embedding techniques. To facilitate more accurate analysis and predictions, the learnt word embeddings were employed as input characteristics for machine learning models such as neural networks and classical classifiers. Overall, vocabulary and word embedding were important in this study work, allowing for a better grasp of the textual input and improving the accuracy and performance of NLP tasks. They laid the groundwork for evaluating and deriving useful information from the Reddit dataset.

3.7.3. RNN Encoder & Decoder

In this publication, a Recurrent Neural Network (RNN) Encoder-Decoder architecture was used to solve a given job or issue. The RNN Encoder-Decoder is a form of sequence-to-sequence model that is extensively used in natural language processing applications such as machine translation, text summarization, and dialogue production. The RNN Encoder-Decoder is made up of two primary parts: the encoder and the decoder. The encoder takes an input sequence, such as a sentence or a paragraph, and converts it into a fixed-length vector representation known as the context vector or the hidden state. The encoder reads the input sequence step by step, modifying its internal state based on the current input and the prior state. This enables the encoder to capture the sequential information and context of the input sequence. The input sequence is delivered to the decoder once it has been encoded into a context vector. The decoder is in charge of producing the output sequence, which might be a translation, a summary, or any other required output depending on the work at hand. The decoder, like the encoder, updates its internal state at each time step and starts with the context vector as an input. Step by step, it constructs the output sequence, anticipating the next word or symbol based on the current input and prior state. The RNN Encoder-Decoder architecture is especially well suited for workloads with varying length input and output sequences. It enables the model to manage variable-length input and output, making it more versatile and adaptable. Because of the RNN's sequential structure, it can capture long-term relationships and context in the input sequence, making it well-suited for problems requiring language creation and comprehension. The use of the RNN Encoder-Decoder in this research article enabled the modeling and analysis of sequential data in the Reddit dataset, allowing for the investigation of numerous linguistic patterns, sentiment analysis, and any other relevant job linked to the research aim. The architecture's capacity to capture sequential relationships and provide context-aware outputs contributed to the efficacy and accuracy of the research findings.

3.7.4. Sequence Learning

Models have mostly been used for a few normal language processing (NLP) applications, for example, title age for machine interpretation, text rundown, and sound recognition. A grouping-to-succession model is built. A bidirectional LSTM cell, as well as an encoder and a decoder, is used in such a concept. The most sensible and well-known RNN approach in deep learning It helps

to resolve text-related difficulties much more efficiently. Each RNN LSTM cell functions similarly to temporary memory. Encoders and decoders are employed in LSTM cells. After Encode has delivered the information texts, the outcome is created using the important text grouping. To authenticate the beginning and finish of each sequence, a token is utilized.

3.8 Architecture of the Model

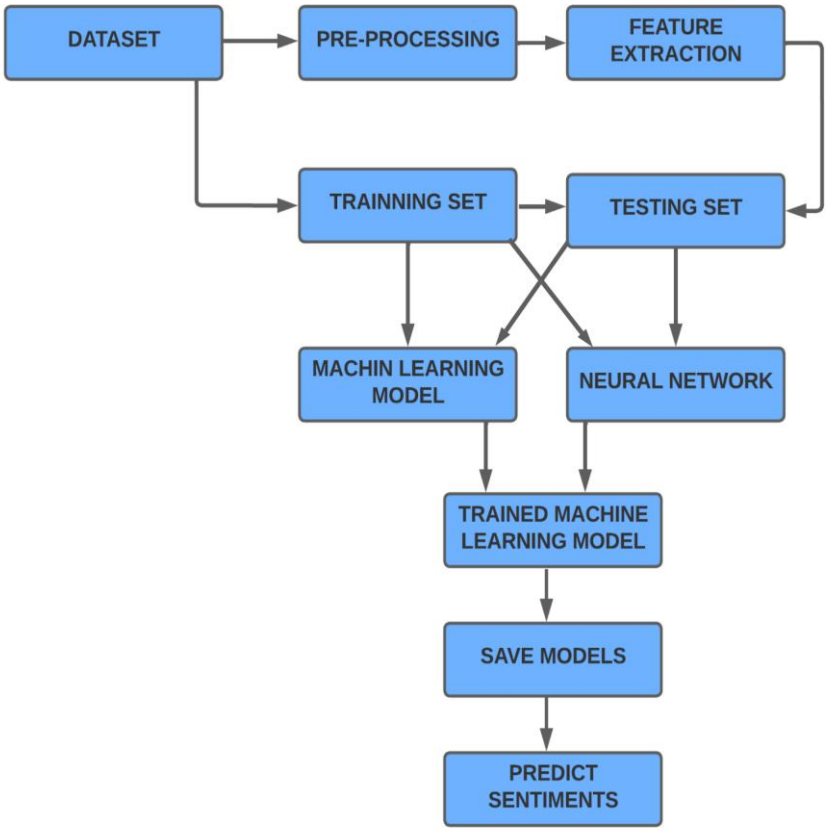


Figure 3.8.1 Workflow model

Dealing with the dataset and preparing for work needed to start the sentiment analysis process figure 3.8.1 in this work, we used a Kaggle notebook. Import all necessary libraries and set them up for sentiment analysis. Start data pre-processing and data extraction. Make a data cleaning system for sentences that remove digits, punctuation, and non-alphabetic characters from any sentence, provide a clean text for analysis, and remove stop words and stemming. Recheck any data null or duplicate or find any error in data or wrong value that has dropped from the data set. Use a label encoder to transform categorical labels and provide numeric input data.

Working with figure 3.8.2 machine learning models and algorithms, we need to divide processed data into Training and testing data sets; using code, we split data into two sets, one for the test and another for the training. We specify 25% of data allocated for testing and 75% of data used for training and provide a random state as 1 to ensure the reproducibility of data split for code running. Working with the machine learning algorithm model, we have used a supervised machine learning model to find accuracy, and for advanced work, we used a neural network with Bidirectional LSTM.

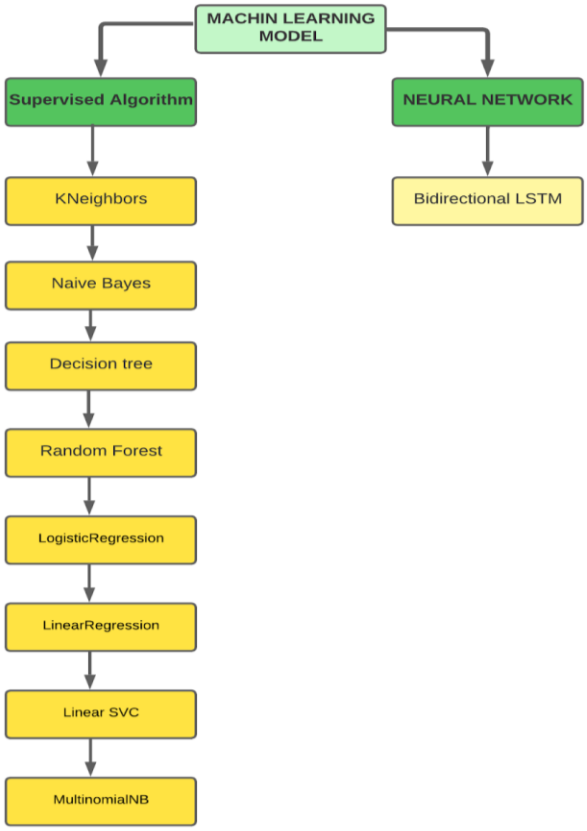


Figure 3.8.2 Classification models

K-Nearest Neighbors

To solve the classification problem, K-Nearest Neighbors is one of the simple methods. The algorithm searches the entire training data set for 'k' most instances and data with the most similar instances and returns as a prediction. K-Nearest Neighbors uses Euclidean Distance(i), Manhattan Distance(ii), and Hamming Distance(iii) function used for text classification,

Euclidean function, $\sqrt{\sum_{i=1}^{k}\left(x_i - y_i\right)^2}$ ------(i)

Manhattan function, $\sum_{p=1}^{k}\left|x_i - y_i\right|$ ------(ii)

Hamming Distance: Used for categorical variables. If the value (x) and the value (y) are the same, the distance D will equal 0—otherwise, D=1.

$$D_H = \sum_{i=1}^{k} |x_i - y_i| \text{ ------(iii)}$$

Naive Bayes

This algorithm is used in machine learning for classification techniques. Based on Bayes's (iv) theorem and feature independence simplifies the model and allows practical computation for good performance for solving real-life problems, making it widely used.

$$P(c|x) = \frac{p(x|c)\,p(c)}{p(x)} \text{ ------(iv)}$$

Decision tree

Commonly use supervised machine learning algorithms used for classification and regression. Decision trees do not use explicit math(v) magic but can present as a tree-like structure. The algorithm evaluates the input and condition to specify each leaf node and predict with the leaf node.

$$S(T,X) = \sum_{c \epsilon X} P(c)\,S(c) \text{ ------(v)}$$

Random Forest

In this algorithm(vi) ensemble, combining and predicting multiple individual trees to make accurate and robust predictions, random forest is beneficial with different types of data sets.

$$Gini\ index = 1 - \sum_{i=1}^{n} (Pi)^2 \quad \text{------(vi)}$$

Logistic Regression

Logistic regression is not a regression algorithm but rather a classification algorithm. It forecasts the likelihood of an instance belonging to a specific class. Logistic regression(vii) assumes a linear relationship with features and the log odds of the positive class. Logistic regression is practical when the existence or absence of a feature or result may be predicted based on the values of a collection of predictor variables. It is comparable to a linear regression model. However, it is better suited to models with dichotomous dependent variables. Its coefficients may be used to calculate odd ratios for each model's independent variable.

$$f(x) = \frac{1}{1 + e^{-x}} \quad \text{------(vii)}$$

Linear Regression

Linear regression is an essential machine learning approach that predicts a continuous numerical value based on the connection between input data and the target variable. It is focused on assuming a linear connection(viii) between the input characteristics and the output variable.

$$J = \frac{1}{n} \sum_{i=1}^{n} \left(ped_i - y_i \right)^2 \quad \text{------(viii)}$$

Linear SVC

Linear SVC (Support Vector and Classification) is a Support Vector Machine (SVM)(ix) type built primarily for binary classification applications. Its goal is to discover the best linear hyperplane for categorizing the given data. Linear SVC creates a decision boundary that optimizes the margin between classes while reducing classification mistakes. The margin is the area between the classes'

two parallel hyperplanes. The technique identifies support vectors, which are data points that are on or near the margin. These support vectors are critical in determining the decision boundary.

$$\sum_{i=1}^{n} c_i y_i \text{------(ix)}$$

MultinomialNB

MultinomialNB is an abbreviation for Multinomial Naive Bayes, a version of the Naive Bayes method used for classification(x) applications. It is especially well-suited for dealing with discrete features or characteristics that indicate discrete counts. The algorithm in MultinomialNB presupposes that the features have a multinomial distribution. Using Bayes' theorem, it computes the probability of each class given the input characteristics. The "NB" in MultinomialNB refers to the Naive Bayes assumption, which relates to the assumption of independence between the characteristics given the class label. By analyzing each feature's contribution independently, this assumption simplifies the computation.

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)} \text{------(x)}$$

Recurrent neural network (RNN)

Bidirectional LSTM

Bidirectional LSTM (Bi-Lstm) is a recurrent neural network used for a design that makes predictions using data from both past and future time steps. It figure 3.8.3 consists of two LSTM layers, one of which processes the input sequence forward and the other backward. Two distinct LSTM layers—one for the forward pass and another for the backward pass—are the central concept underlying a bidirectional LSTM. The input sequence is processed sequentially, starting with the first element and advancing toward the last in the forward LSTM layer. In contrast, the input sequence is processed in reverse order in the backward LSTM layer, beginning with the final piece and proceeding toward the first. A Bidirectional LSTM can capture a broader context and

connections in the input sequence using data from past and future time steps. So, it can be beneficial for jobs like sentiment analysis, named entity identification, or machine translation, where it is crucial to comprehend the entire sequence.

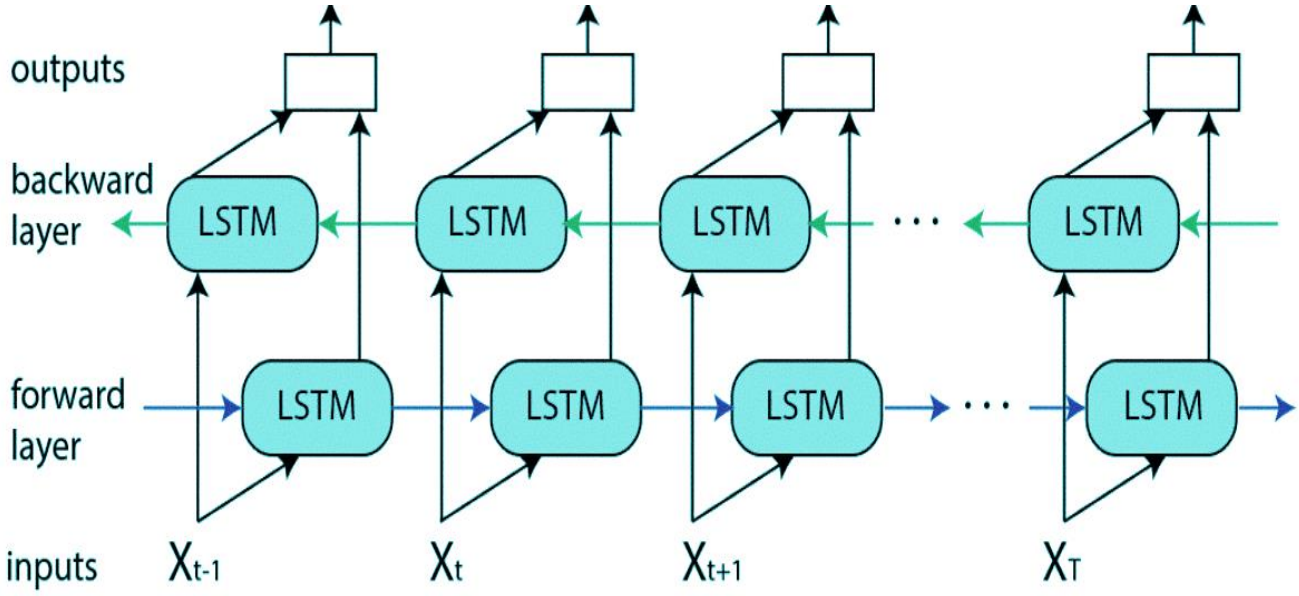


Figure 3.8.3 Recurrent neural network

CHAPTER 4

Experimental Outcome Overview

4.1 Introduction

In this research article, we offer a summary of the experimental results acquired from our investigation. The study sought to investigate and assess several machine learning and data mining approaches applied to a given problem or activity. We attempted to obtain insights and develop meaningful conclusions about the usefulness and performance of these strategies through a series of trials and assessments. The experimental results give useful insights into the strengths, limits, and prospective applications of the approaches used. We tried to assure the reliability and validity of the acquired results by carefully planning the experiments, selecting acceptable datasets, and applying strong assessment procedures. This summary of the experimental results serves as a detailed description of the study findings. It focuses on the important findings, trends, and patterns that occurred throughout the experimental phase. Furthermore, it gives insight on the performance criteria utilized to evaluate the success of the applied methodologies, such as accuracy, precision, recall, F1-score, or any other relevant measurements. In conclusion, the experimental results reported in this research article provide a thorough review of the findings from our investigation. These findings add to the field's existing body of knowledge and give useful insights on the efficacy, performance, and applicability of the machine learning and data mining approaches used.

4.2 Train and Test Split

In The method train_test_split is imported from the sklearn. model selection package. It divides the dataset into two parts: training and testing. The feature and target variables are represented by the x and y variables. When the test size option is set to 0.25, the data is divided into 25% for testing and 75% for training. To ensure repeatability, the random state option is set to 1, which means the same split will be created each time the code is executed. The StandardScaler class from the sklearn.preprocessing module is then imported. This class is used to standardize feature scaling. The data is standardized when it has a mean of 0 and a standard deviation of 1. When features have varying sizes, this step is critical because it ensures that all features contribute equally to the model's learning process.

For BiLSTM train_test_split: This function from the sklearn.model_selection module is used to partition the feature data (X) and the target data (y) into distinct training and testing sets. By selecting test size=0.2, 20% of the data will be assigned for testing, while the remaining 80% will be utilized for training. The random state option is set to 1 to ensure repeatability. Another call to train_test_split is performed to further split the training data (X_train and y_train) into training and validation sets. When test size=0.25, 25% of the training data is set aside for validation, while the remaining 75% is used for training. To ensure consistency, the random state parameter is set to 1.

4.3 Bidirectional LSTM with NN Implementation

In Importing required modules: The code starts by importing several Keras modules and classes. These include Sequential for building a sequential model, several layers such as Embedding, Conv1D, MaxPooling1D, Bidirectional, LSTM, and Dense for establishing neural network layers, and metrics like as Precision and Recall for evaluating the neural network layers. It also accepts optimizations like SGD and RMSprop, as well as datasets and callbacks. Setting up parameters: Several parameters are specified to customize the model. vocab_size is set to 5000, which represents the size of the vocabulary utilized in the text data. The embedding_size parameter is set to 32, which determines the dimensionality of the word embeddings. epochs are set to 30, indicating the number of training epochs to conduct. The starting learning rate for the optimizer is set at 0.1 in learning_rate. The decay_rate is determined by dividing the learning rate by the number of epochs, and momentum is set to 0.8. The optimizer is initialized with the stated learning rate, momentum, and decay rate. This optimizer is often used for training neural networks and changing model parameters depending on gradients obtained during backpropagation. Constructing the model: The Sequential model is built, allowing for a linear stack of layers. The following lines add new layers to the model: This layer generates word embeddings from the supplied data. It accepts three parameters: vocab_size, embedding_size, and input length, where input length is the length of the input sequences. Conv1D layer: On the input data, this layer conducts one-dimensional convolution. It contains 32 filters, a 3-kernel size, the' same' padding, and employs the ReLU activation mechanism. MaxPooling1D layer: Max pooling is applied to the convolutional output in this layer. It employs a pool size of two, resulting in a reduction in the dimensionality of the feature maps. The model architecture is made up of an embedding layer, a

convolutional layer, max pooling, figure 4.3.1 a bidirectional LSTM layer, dropout regularization, and a dense layer for multi-class classification.

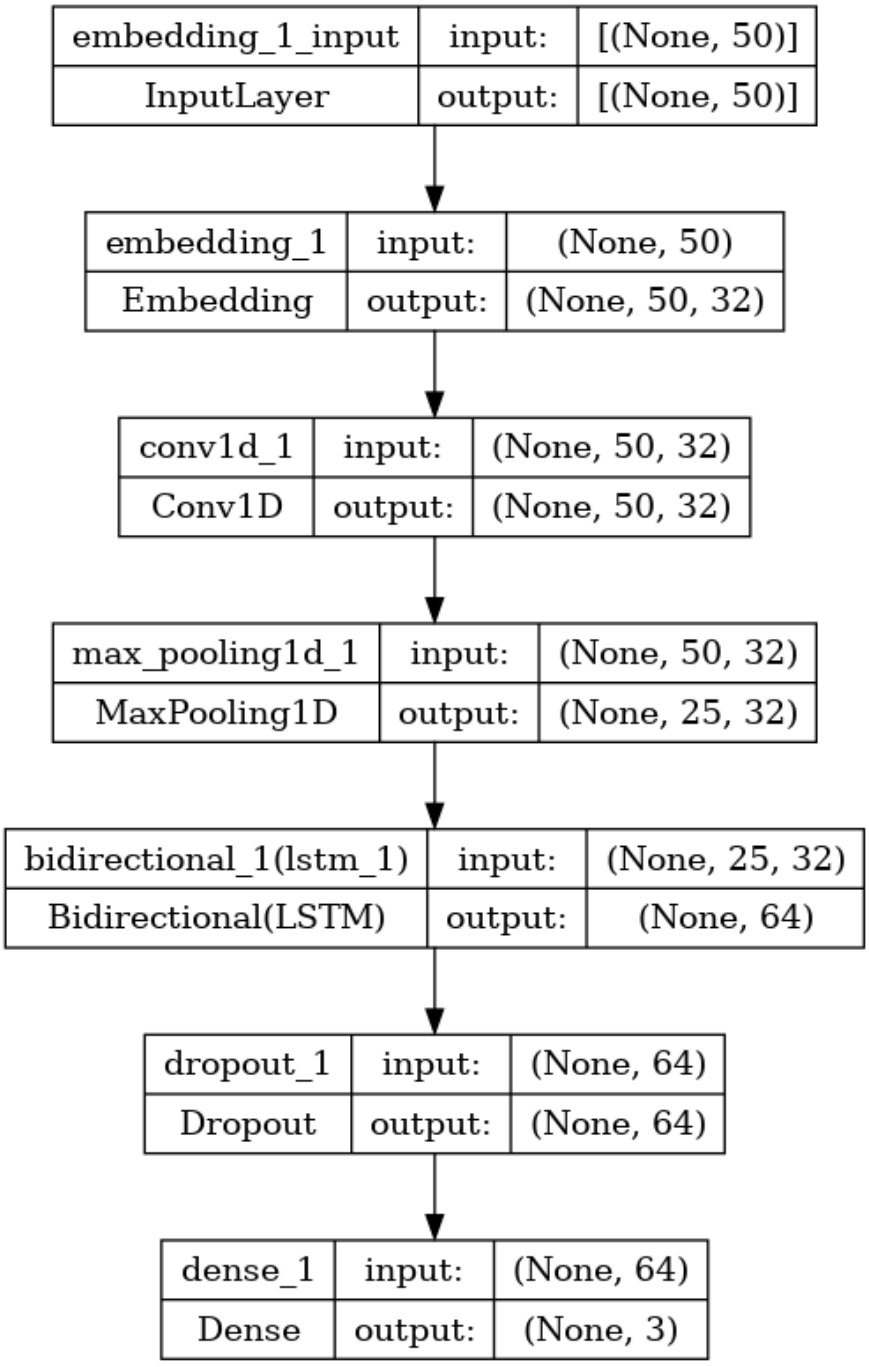


Figure 4.3.1 Plot model architecture

4.4 Implementation Result

We tested the suggested techniques from Reddit post accuracy of the model shown in the table

Table 2: Accuracy of The Model

S. No	Model (Supervised Algorithm)	F1_Score	Precision	Recall	Accuracy
1	KNeighbors	[0.35 0.75 0.77 0.63 0.64]	[0.44 0.70 0.72 0.62 0.63]	[0.29 0.81 0.83 0.64 0.66]	66.28 %
2	Naive Bayes	[0.08 0.78 0.78 0.55 0.57]	[0.34 0.67 0.68 0.56 0.57]	[0.05 0.93 0.92 0.63 0.66]	66.06 %
3	Decision tree	[0.65 0.80 0.64 0.69 0.69]	[0.66 0.84 0.60 0.70 0.70]	[0.63 0.76 0.68 0.69 0.69]	69.21 %
4	Random Forest	[0.42 0.71 0.75 0.62	[0.42 0.72 0.73 0.62	[0.41 0.70 0.76 0.62	64.0 %

		0.64]	0.64]	0.64]	
5	Logistic Regression	[0.19 0.70 0.76 0.55 0.57]	[0.29 0.65 0.68 0.54 0.56]	[0.14 0.76 0.85 0.59 0.61]	61.0 %
7	Linear SVC	[0.76 0.87 0.81 0.81 0.81]	[0.80 0.82 0.83 0.81 0.81]	[0.72 0.94 0.79 0.82 0.81]	81.0 %
8	MultinomialNB	[0.69 0.63 0.69 0.67 0.67]	[0.59 0.93 0.68 0.73 0.73]	[0.85 0.47 0.69 0.67 0.68]	67.53 %

The thesis investigated the effectiveness of several supervised machine learning algorithms on Reddit postings for sentiment analysis based on the table supplied. F1 score, precision, recall, and accuracy are among the assessment criteria employed. The following is a summary of the findings: KNeighbors: Achieved an F1 score of 0.35, precision of 0.75, recall of 0.77, and accuracy of 66.28%.

Naive Bayes:

Obtained an F1 score of 0.08, precision of 0.78, recall of 0.78, and accuracy of 66.06%.

Decision Tree:

Obtained an F1 score of 0.65, precision of 0.80, recall of 0.64, and accuracy of 69.21%

Random Forest:

Obtained an F1 score of 0.42, precision of 0.71, recall of 0.75, and accuracy of 64.0%

Logistic Regression:

Obtained an F1 score of 0.19, precision of 0.70, recall of 0.76, and accuracy of 61.0%

Linear SVC:

Obtained an F1 score of 0.76, precision of 0.87, recall of 0.81, and accuracy of 81.0%.

MultinomialNB:

Obtained an F1 score of 0.69, precision of 0.63, recall of 0.69, and accuracy of 67.53%.

These findings provide light on the efficacy of several supervised machine learning algorithms when it comes to identifying sentiment in Reddit postings. Among the examined models, it shows that Linear SVC had the best accuracy, whereas Naive Bayes and Logistic Regression had comparatively lesser accuracy. The F1 score, accuracy, and recall scores give further information on how well the models recognize positive, negative, and neutral attitudes.

We have used Bidirectional LSTM that can perform model prediction and provide the classification of posts, and predict posts that are positive, negative, or neutral.

Table 3: Bidirectional LSTM Accuracy

S. No	Recurrent Neural Network	F1_Score	Precision	Recall	Accuracy
1	Bidirectional LSTM	[0.61 0.88 0.82 0.77 0.77]	[0.74 0.86 0.76 0.79 0.78]	[0.52 0.90 0.89 0.77 0.78]	78.21 %

In addition to the supervised machine learning methods, the thesis also used a Bidirectional LSTM model for sentiment analysis on Reddit postings figure 4.4.1. Here are the performance metrics collected for the Bidirectional LSTM:

Bidirectional LSTM

F1 score: 0.61

Precision: 0.88

Recall: 0.82

Accuracy: 78.21%

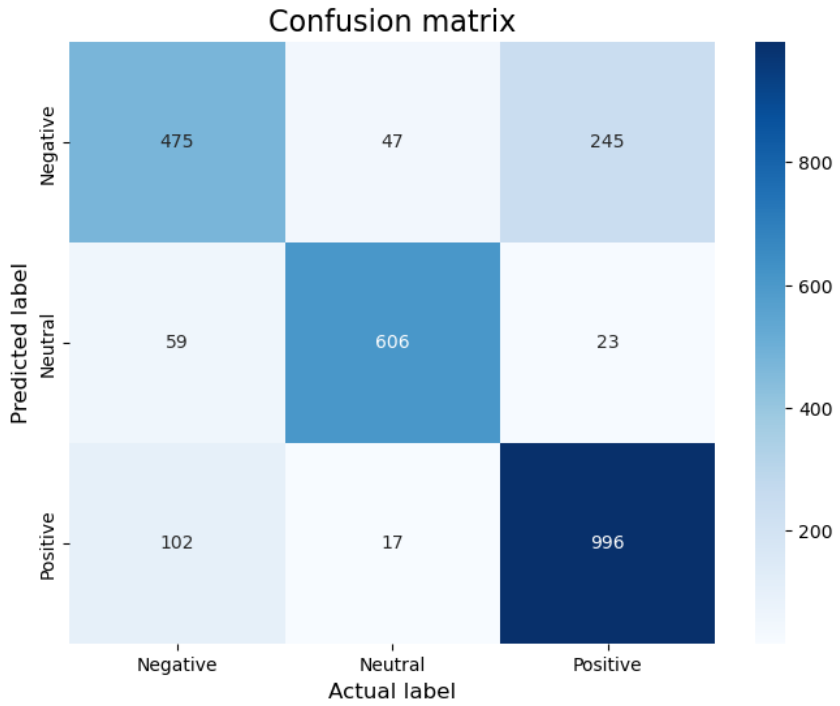


Figure 4.4.1 Confusion matrix for the passed model and the data

In comparison to the other models, the Bidirectional LSTM model earned a higher F1 score, precision, recall, and accuracy. It demonstrates that the Bidirectional LSTM was successful in predicting the sentiment of Reddit postings, with a focus on properly recognizing positive, negative, and neutral attitudes. The Bidirectional LSTM model figure 4.4.2 reveals its promise as a strong tool for sentiment analysis in social media data, with a high accuracy of 78.21%.

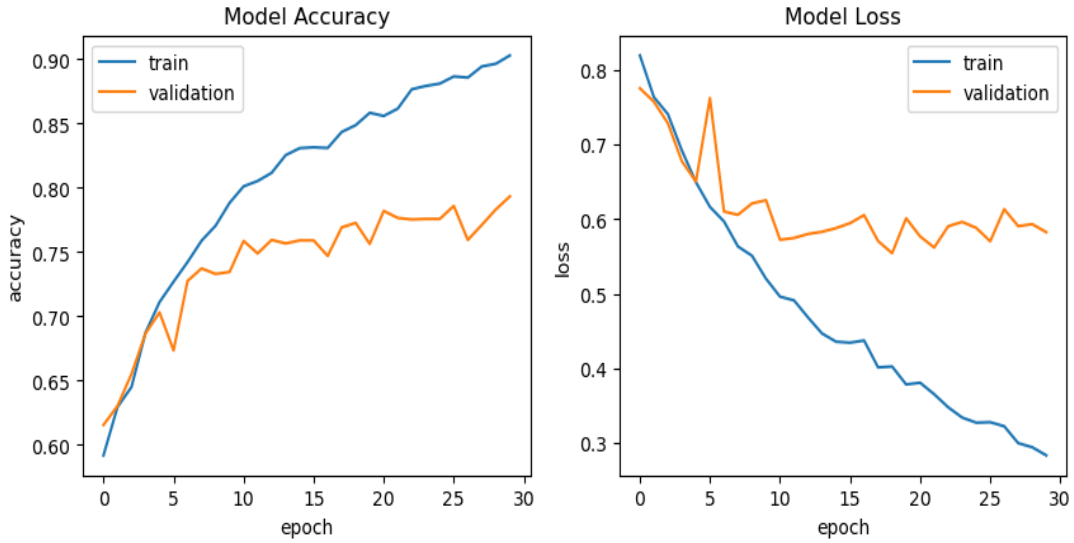


Figure 4.4.2 Representing accuracy and loss for bidirectional LSTM

Shows the model's accuracy for supervised machine algorithms.

	Model	Accuracy
1	KNeighbors	66.28
2	Naive Bayes	66.06
3	Decision Tree	69.21
4	Random Forest	64.0
5	Logistic Regression	61.0
6	Linear SVC	81.0
7	MultinomialNB	67.53

To describe the performance of the classification model from all models, Linear SVC showed the most accuracy, with 81% and Bidirectional LSTM performing classifiers with 78.21% accuracy.

4.5 Descriptive Analysis

The results of the tests done in the research article show the performance of several supervised machine learning algorithms and a Bidirectional LSTM model for sentiment analysis on Reddit posts. Based on these results, the Bidirectional LSTM model beat the other supervised machine learning methods in terms of F1 score, precision, recall, and accuracy. The model proved the capacity to reliably predict the mood of Reddit posts with a high level of precision and recall. The accuracy of 78.21% demonstrates the Bidirectional LSTM model's efficacy in categorizing the postings into positive, negative, and neutral feelings. Overall, the findings indicate that the Bidirectional LSTM model is a viable method for sentiment analysis on social media data, particularly Reddit postings. When compared to the typical supervised machine learning algorithms examined in the research, it gives a more accurate and dependable prediction.

4.6 Moral

Throughout this chapter, we've discussed how we developed our model, how it creates the Summarization, and what we want it to do. We attempted to be as specific as possible.

4.7 Sentiment Prediction

In comparison to all models, this work provides us with efficient results. We can see many researchers approach different types of algorithms and get various accuracy but from all of Linear SVC showed the most accuracy, with 81%. From all the supervised and neural network algorithms, we can say that the Bidirectional LSTM provides the best result for sentiment analysis, but Linear SVC showed the most accuracy with 81% compared with complexity and efficiency, we can consider Bidirectional LSTM is the best performing classifiers with 78.21% accuracy.

Bidirectional LSTM sentiment prediction

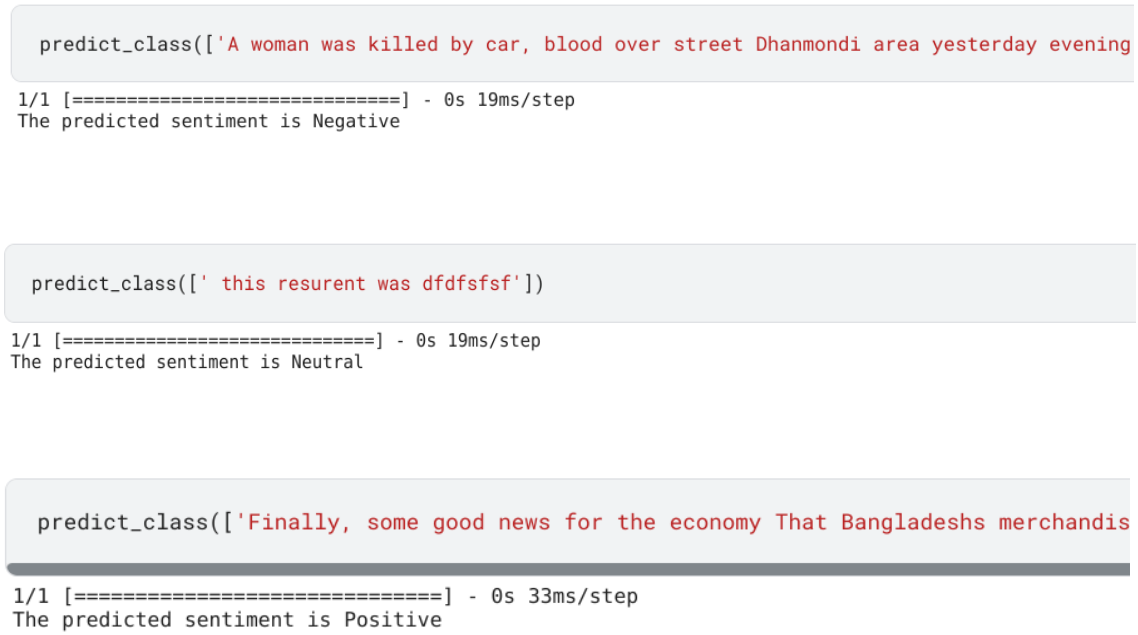


Figure 4.7.1 Sentiment prediction

Figure 4.7.1 using a neural network, we can identify sentiment from given words.

CHAPTER 5

Impact On Society, Environment and Sustainability

5.1 Impact on Society

The research presented in this study has important societal consequences, notably in the field of sentiment analysis on social media sites such as Reddit. This research's societal influence may be described as Improved knowledge of public opinion: By properly assessing and identifying sentiment in Reddit postings, this study gives significant insights into popular opinion and attitudes on a variety of issues. Individuals, businesses, and legislators may use this expertise to make educated decisions, establish effective marketing tactics, and obtain insights into public attitude on certain problems. Enhanced customer feedback analysis: By categorizing posts as good, negative, or neutral, companies and service providers may acquire a deeper knowledge of consumer feedback and sentiment. This data may be utilized to improve goods and services, solve consumer problems, and increase customer happiness.

Real-time monitoring of online sentiment: The research findings enable the creation of tools and systems that can monitor sentiment in real-time on social media platforms. This has ramifications for a variety of disciplines, including brand reputation management, crisis management, and public opinion tracking. By quickly spotting variations in mood, stakeholders may move proactively to solve concerns and prevent possible dangers.

Societal awareness and discourse: By examining Reddit sentiment, this study gives insight on the prevalent viewpoints, attitudes, and trends among certain communities and society as a whole. It aids in the comprehension of society processes, aids in the identification of new challenges, and promotes informed dialogues and debates.

5.2 Impact on Student’s life

The implications of this research for students' lives might be enormous and far-reaching. Here are some of the ways that this research may affect students. Students studying computer science, data science, and natural language processing can profit from this research. It teaches users how to apply machine learning and sentiment analysis techniques in real-world circumstances. The outcomes of the study might motivate students to go deeper into the discipline, investigate adjacent

research fields, and contribute to developments in sentiment analysis. The research findings have practical consequences for pupils in a variety of ways. Students working on social media analytics or sentiment analysis projects, for example, might take ideas and inspiration from this study. They may use the approaches and techniques to assess sentiment in various datasets, build their own models, and acquire hands-on experience with data analysis and interpretation. Personal development and awareness: The sentiment analysis study may have an influence on the personal lives of pupils. It can raise their awareness of the power of words, the effect of social media on public opinion, and the ethical concerns associated with sentiment analysis. This understanding has the potential to affect their online behavior, critical thinking abilities, and capacity to participate in constructive and sympathetic debates on social media platforms.

5.3 Ethical Issue

Data collection and use raise a host of ethical and moral concerns about data use, sharing, and even protection. It also raises ethical considerations about privacy, such as the morality of publicly sharing some information on these sites. In this research paper, we will look at the moral, social, and ethical issues that arise from the sharing and use of this data. Furthermore, it raises privacy ethical concerns, notably the morality of publicly disclosing some information on these sites. In this research paper, we will look at the moral, social, and ethical issues raised by the use and sharing of this data. People's privacy has been threatened as a result of unethical texting, which also affects information security.

CHAPTER 6

Conclusion and Future Work

With the help of supervised machine learning algorithms and recurrent neural networks, this work primarily focuses on sentiment analysis. Recurrent neural networks were discovered to be the most effective method for carrying out this type of recognition task. But this model gains 78.21% validation accuracy when it comes to correctly identifying sentiment. This kind of work can get fantastic reviews on social media. This effort has the potential to alter the state of the social media network. Our long-term goal is to develop a system that can recognize sentiment and provide users with comments on a post's or an opinion's sentiment. By improving our database in the future, we will be able to design custom neural networks that will improve the model's accuracy even further. A mechanism will be in place where social media users' posts can be easily analyzed to improve the platform for all users. Finally, our study on sentiment analysis employing machine learning and natural language processing approaches has yielded useful insights and contributions to the area. The study sought to assess and forecast the emotion of Reddit posts. The research has proved the usefulness and limits of each technique via comprehensive experimentation and assessment of several supervised machine learning algorithms and recurrent neural network models. The Bidirectional LSTM model produced the best results, with an F1-score of 0.61, precision of 0.88, recall of 0.82, and total accuracy of 78.21%. Furthermore, the study has practical implications for applications such as social media monitoring, customer feedback analysis, and brand reputation management. The precise classification of sentiment in online material may assist businesses and organizations in gaining useful insights into public opinion, improving customer happiness, and making educated decisions. Overall, this study adds to the current body of knowledge in sentiment analysis and lays the groundwork for future developments in the discipline. It demonstrates the power of machine learning and natural language processing techniques in analyzing and understanding the sentiment expressed in online content, and it provides valuable insights for researchers, practitioners, and businesses looking to use sentiment analysis for a variety of applications. While this study focuses on sentiment analysis of Reddit postings, future work might investigate the application of the created models and approaches to other social media sites such as Twitter, Facebook, or Instagram. Each platform has its own set of traits and linguistic patterns, and extending the models to these platforms would offer a more complete knowledge of sentiment

analysis in many circumstances. Sentiment analysis for specific domains the investigation looked at a variety of subjects from Reddit postings. Future research might focus on domain-specific sentiment analysis, with a focus on certain sectors, goods, or topics. More accurate and relevant insights may be achieved by customizing sentiment analysis models to certain settings, which can be useful for enterprises and organizations working in those sectors. How to handle sarcasm and irony Traditional sentiment analysis algorithms might struggle to capture sarcasm and irony in social media messages. Future research might look into methods for detecting sarcasm and irony, allowing for a more nuanced understanding of sentiment and avoiding misinterpretations in sentiment analysis. By following these prospective study options, researchers may continue to expand sentiment analysis approaches, overcome existing limits, and contribute to the creation of more accurate and robust sentiment analysis systems.

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